Name ⇒ Deep Punia
Sap id ⇒ 500123744
Batch ⇒ 3 (Data Science)

ASSIGNMENT - 1

Q-1 A scalar LSTM ..... $x_t = 0.5$, with hidden state $h_{t_1} = 0.2$.... Compute f_t, i_t, C_t, $\tilde{C}_t$, O_t & i_t.

⇒ $x_t = 0.5$ $\quad h_{t-1} = 0.2$ $\quad C_{t-1} = 0.4$

$$f_t = \sigma(w_f \cdot x_t + U_b \cdot h_{t-1} + b_b)$$
$$= \sigma[(0.7 \times 0.5) + (0.3 \times 0.2) + 0.1)$$
$$= \sigma(0.5100)$$
$$\therefore \sigma(0.5100) = \frac{1}{1 + e^{(-0.5100)}} = \frac{1}{1.6005} = \boxed{0.6248}$$

$$\therefore \boxed{f_t = 0.6248}$$

$$i_t = \sigma(w_i \cdot x_t + U_i \cdot h_{t-1} + b_i)$$
$$= \sigma((0.6 \times 0.5) + (0.4 \times 0.2) + 0.0)$$
$$= \sigma(0.3800) = \frac{1}{1 + e^{-(0.3800)}} = \frac{1}{1.6839}$$
$$= 0.5939$$

$$\therefore \boxed{i_t = 0.5939}$$

$$C_t = \tanh(w_c \cdot x_t + U_c \cdot h_{t-1} + b_c)$$
$$= \tanh((0.9 \times 0.5) + (0.2 \times 0.2) + 0.0)$$
$$= \tanh(0.4900)$$
$$\tanh(0.4900) = \frac{e^2 - e^{-2}}{e^2 + e^{-2}} = \frac{e^{(0.4900)} - e^{(-0.4900)}}{e^{(0.4900)} + e^{(-0.4900)}}$$
$$= 0.4542$$

$$\therefore \boxed{\tilde{C}_t = 0.4542}$$

$$o_t = \sigma(W_o x_t + U_o \cdot h_{t-1} + b_o)$$
$$= \sigma((0.5 \times 0.5) + (0.5 \times 0.2) + 0.1)$$
$$= \sigma(0.4500)$$

$$\sigma(0.4500) = \frac{1}{1+e^{-(0.4500)}} = 0.6106$$

$$\therefore \boxed{o_t = 0.6106}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$
$$= (0.6248 \times 0.4) + (0.5939 \times 0.4542)$$

$$\boxed{c_t = 0.5197}$$

$$h_t = \tanh(c_t) \odot o_b$$
$$= \tanh(0.5197) \times 0.6106$$
$$h_t = 0.4775 \times 0.6106$$

$$\boxed{h_t = 0.2916}$$

Q-2 (a) Using the standard ....., how many trainable parameters does the layer have?

⇒ Input Dimension (x_{dim}) = 10

Hidden units (h_{dim}) = 32

For a single gate, linear transformation is:

$$gate = \sigma(Wx_t + Uh_{t-1} + b)$$

⊙ Input Weight: Parameters = $h_{dim} \times x_{dim}$
$= 32 \times 10 = 320$

⊙ Recurrent / Hidden Weight: Parameters = $h_{dim} \times h_{dim}$
$= 32 \times 32 = 1024$

⊙ Bias: One bias: per hidden unit
parameters = $h_{dim} = 32$

Total Trainable Parameters = $320 + 1024 + 32$
$= 1376$

∵ We have 4 gates.

∴ Total Trainable Parameters = $4 \times 1376 = \boxed{5504}$

(b) Framework ....., How many parameter does the layer have under that convention.

⇒ In such frameworks, the transformation for each gate includes two independent bias - one for input projection & one for hidden state projection.

- Input weight = $32 \times 10 = 320$
- Hidden weight = $32 \times 32 = 1024$
- Input Bias = 32
- Hidden Bias = 32

Total parameters per gate = $320 + 1024 + 32 + 32 = \boxed{1408}$

∴ Total Trainable parameters = $4 \times 1408 = \boxed{5,632}$

Q3 (a) Compute scaling ....., after $k = 50$ steps.

⇒ for $k = 10$ steps

Scaling factor = $0.9^{10} \approx \boxed{0.3487}$

for $k = 50$ steps

Scaling factor = $0.9^{50} \approx \boxed{0.0052}$

(b) For $k = 50$, compute ..... for $f = 0.99$.

⇒ for $f = 0.5$

Scaling factor = $0.5^{50} \approx \boxed{0.882 \times 10^{-15}}$

for $f = 0.99$

Scaling factor = $0.90^{50} \approx \boxed{0.6050}$

(c) Explain the result ?

⇒ ⊙ Smaller forget gate values causes the gradient to vanish quickly over 50 steps, which makes learning long-term dependencies nicely impossible.

⊙ LSTM solve this by learning to keep the forget gate value close to 1.0 when memory needs to be preserved, which creates a direct linear highway for the gradient to flow backwood through the cell state.

⊙ Vanilla RNN lack this path & multiply gradients through repeating non-linear functions, causing them to vanish completely.

Q-4

(a) Total Hourly Records = Years × Day/Year × Hour/day

= 3 × 365 × 24

= **26280**

(b) Total Input/Output Window:

Total Window Size = 168 + 24 = 192 hrs
(lookback) ↓ (prediction)

⇒ Total Records − Total window size + 1

= 26280 − 192 + 1

Total Windows = 26089

(c) With Batch size = 64

Shape format = (Batch size, Time step, Features)

Input Batch Shape = (64, 168, 3)

output Batch Shape = (64, 24, 1) { If predicting only 1 feature }

If predicting every feature = (64, 24, 3)

(d) Training Records = 70% of 26280

$$= \frac{70}{100} \times 26280$$

$$= \boxed{18,396 \text{ records}}$$

Training Windows = 18396 − 192 + 1

= 18,205 windows

Mini Batches per Epoch = $\frac{18205}{64}$

= 284.45

≈ 284 (including partial batch)

(e) Parameter for 1 Gate :

- Input Weights : 64 × 3 = 192
- Recurrent weight : 64 × 64 = 4096
- Bias : 64

Total Per gates : 192 + 4096 + 64

$\Rightarrow \boxed{4352}$

∵ 4 Gates are there

∴ Total Trainable Parameters = 4 × 4352 = $\boxed{17,408}$

Q-5

(a) Min-Max Normalization = $\frac{x - x_{min}}{x_{max} - x_{min}}$

Normalised Value = $\frac{95-20}{120-20} = \frac{75}{100} = \boxed{0.75}$

(b) The O/P probability = 0.83

Threshold = 0.5

∵ O/P > Threshold, so model predicts a failure (Negative class)

(c) True Positive (TP) = 30 (correctly flagged failure)
False Positive (FP) = 45 - 30 = 15 (false flagged failure)
False Negative (FN) = 40 - 30 = 10 (missed actual failure)
True Negative (TN) = 1000 - 30 - 15 - 10 = 945 (correctly flagged normal)

$$\therefore \text{Precision} = \frac{30}{45} = 0.667 \approx \boxed{66.7\%}$$

$$\text{Recall} = \frac{30}{40} = 0.75 \approx \boxed{75\%}$$

$$\text{Accuracy} = \frac{30+945}{1000} = \frac{975}{1000} = 0.975 = \boxed{97.5\%}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \frac{0.667 \times 0.750}{(0.667 + 0.750)} = 0.7061 \approx \boxed{70.61\%}$$

(d) The team should prioritise Recall, missing a real pump failure causes costly downtime / catastrophic damage.

Accuracy is misleading here due to severe class imbalance, only 4% of the total dataset contain actual failures. A broken model predicting "Not failure" every time still gets a high 96% accuracy.